

Vishnu Sundhan

Software Engineer | GenAI & Agentic Systems | LLM Infrastructure

linkedin.com/in/vizen0 | portfolio: zenodium.github.io/mysite

Applied AI Engineer with a background in ML/DS who loves to learn and solve ambiguous challenges from scratch, having over 1.5 years of cumulative, hands-on experience across 5 internships, specializing in RAG architectures, agentic workflows, observability and optimization of agents, and deploying scalable backend solutions in Python. 7x Hackathon Wins for rapidly prototyping microservices and multi-agent systems across competitive events (including the MIT Hackathon, ML Olympiad Winner - Kaggle Competition, Montclair Hackathon, 3 Percent Hackathon, and Stevens Hackathon).

EDUCATION

New York University

M.S. in Computer Science | GPA: 3.78

New York, USA
Aug 2024 – May 2026

Rajalakshmi Engineering College, Anna University

B.E. in Computer Science and Engineering | GPA: 8.94/10

Chennai, India
Nov 2020 – May 2024

EXPERIENCE

Runova Inc

Software Engineer Intern

New York, USA
Jan 2026 – May 2026

- Developed a fully offline Android Skincare application in Kotlin by migrating a Flask REST API to native mobile, eliminated cloud dependency entirely, cut server-round-trip latency by 100%, and improved runtime stability and deployment reliability across distributed edge environments.
- Implemented LLM observability to track token cost and latency, used Veris AI's GCP-integrated simulation sandbox to stress-test and certify agents against accuracy, cost, and latency SLAs before customer deployment, partnering with stakeholders on go-live requirements.
- Engineered a real-time on-device processing pipeline using CameraX and Kotlin coroutines for facial analysis, designed an asynchronous parallel execution model supporting simultaneous computer vision inference, ML execution, and UI rendering within sub-100ms latency constraints.

Tata Technologies

Software Engineer Intern (Gen AI)

Pune, India
Jun 2025 – Jul 2025

- Designed and deployed multi-agent microservices using LangGraph, CrewAI, and FastAPI with a React/TypeScript frontend for context-aware project timeline coordination and a privacy-first cross-project meeting Assistant with ReAct-style reasoning for enterprises, reducing project ETA by 2 weeks.
- Fine-tuned PyTorch models for industrial machinery anomaly detection for pipe welding, achieving 96% precision and designed a live monitoring dashboard with optimized query caching that reduced diagnostic review time by 30%+, enabling faster fault identification on real time manufactured pipes.

Populus Empowerment Network

Software Developer Intern

Chennai, India
Jan 2024 – Jun 2024

- Built fault-tolerant ETL pipelines from ingesting various web sources and cleaning data to dashboard visualizations for processing 6M+ bilingual electoral records using Python, SQL, and Selenium. Maintained 98%+ data consistency and reliable downstream analytics workflows for stakeholders.
- Automated data extraction from ambiguous scanned documents using custom-trained CNNs, Coordinate Correction, Tesseract OCR, and Amazon Textract, which reduced manual processing effort by 50%+ and cut extraction error rates significantly across multiple document formats.

Rajalakshmi Engineering College

AI Research Assistant

Chennai, India
Oct 2023 – Jan 2024

- Published IEEE paper on deep learning-based Alzheimer's detection from MRI data, achieving 99.06% accuracy via a TensorFlow/PyTorch ensemble with custom OpenCV preprocessing and multi-stage feature extraction.
- Co-authored IEEE paper on brain MRI multi-class segmentation using CNN and Hugging Face transformer pipelines; validated classification on real-world clinical imaging datasets with high region detection accuracy.

LogicField Technologies Pvt Ltd

Machine Learning Intern

Chennai, India
Jun 2023 – Aug 2023

- Developed a geospatial risk assessment ensemble model (XGBoost, LightGBM) in Python utilizing GeoPandas and MLflow for optimizing insurance taxes, processing over 500,000 historical records to accurately forecast location-specific disaster probabilities (earthquakes, tsunamis) with 89% accuracy.
- Conducted comprehensive data analysis to provide actionable recommendations that reduced potential financial exposure for enterprise clients. Designed and deployed an interactive Node.js and Express dashboard to serve real-time risk metrics, reducing manual reporting time by 40%.

PROJECTS

The Bitcoin Quorum – Autonomous Multisig AI Treasury

Apr 2026

LangGraph, LangChain, FastAPI, ChromaDB, Bitcoin Miniscript, Groq, Vapi, Next.js, Python

- Developed multi-agent autonomous Bitcoin treasury using LangGraph with 3-of-4 Miniscript multisig enforcement; agents independently think, vote, and cryptographically co-sign, making unilateral fund movement mathematically impossible – MIT Bitcoin Hackathon Community Pool Winner.
- Orchestrated parallel fraud detection across Tavily vendor search, Bitcoin Abuse blacklists, WHOIS domain screening, and Blockstream address analysis; integrated ChromaDB RAG for live policy retrieval and a Vapi voice interface for real-time treasury commands using Next.js as the interface.

Mar 2026

Shadow Me - Privacy-Preserving Personal AI Agent

LLMs, RAG, ReAct, Vertex AI, Presidio, BLIP, Whisper, ChromaDB, Telegram API, Python

- Built a conversational RAG AI agent with ReAct-style reasoning on 10K+ Telegram messages; integrated Microsoft Presidio to detect and redact 12 PII entity types before LLM inference, ensuring zero private data exposure prior to Vertex AI model serving.
- Engineered an offline multimodal pipeline processing images via BLIP captioning, OCR, and voice notes via Whisper transcription; converted all media to semantic embeddings for RAG retrieval, reducing raw data sent to the LLM by 90%+ versus direct transfer.

Nov 2025

Refractor Radar – AI Code Analysis Agent

TypeScript, Semantic Embeddings, AST Parsing, ChromaDB, CrewAI, NetworkX, Gemini API

- Built an AI-powered codebase analysis backend extracting AST structures and embedding functions via Gemini API; identified code duplication through semantic similarity clustering, enabling automated refactoring recommendations. Exposed the service via a Node.js web interface.

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, R, Kotlin, HTML/CSS

Backend & APIs: DSPy, FastAPI, Flask, REST APIs, Microservices, Distributed Systems, Node.js, React.js, LangGraph, CrewAI, LangChain, Google ADK, AutoGen, PydanticAI, Docker, Git, Linux, CI/CD, Caching, System Design, Redis, Ollama

Frontend & Data: React, PowerBI, Pandas, NumPy, Matplotlib, Plotly, Jupyter, D3.js, GraphRAG, Hadoop, Apache Spark

ML & MLOps: PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, Hugging Face, Transformers, MLflow, Prompt Engineering, Fine-Tuning, NLP

Cloud & Databases: GCP, Vertex AI, AWS SageMaker, AWS S3, DynamoDB, ChromaDB, Gemini API, Amazon Textract